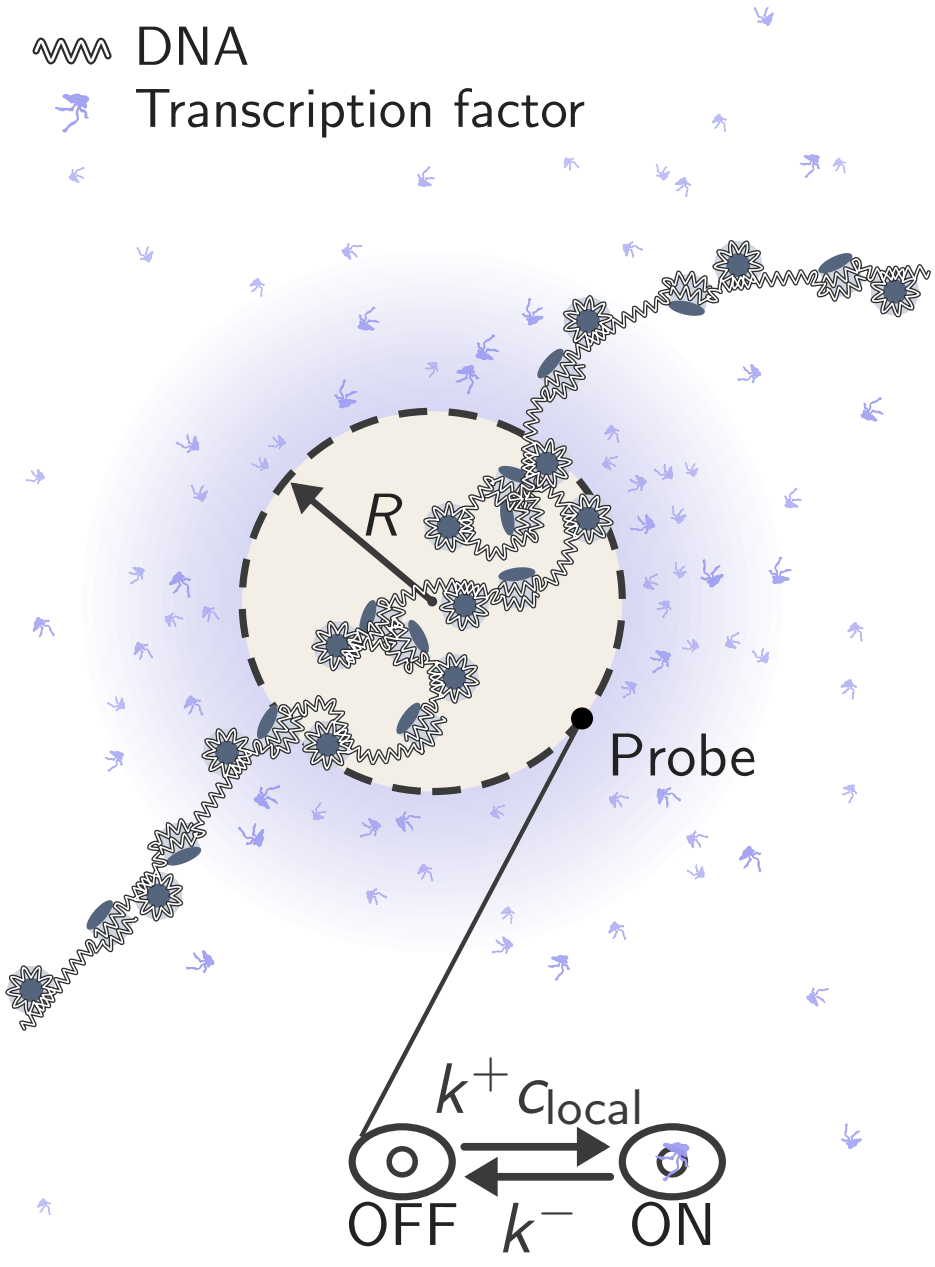


Free probe

$$\dot{\mathcal{F}}_{\text{free}}(c) = \frac{1}{\tau_r(c)}$$

Probe on the anchor

$$\dot{\mathcal{F}}(c) = \underbrace{\frac{1}{\tau_r(\phi)}}_{\text{probe}} \underbrace{\left(\frac{d \log \phi}{d \log c}\right)^2}_{\text{surface}}$$

τ_r mean cycle time of the probe
ϕ TF volume fraction at the probe